ASSIGNMENT/ASSESSMENT ITEM COVER SHEET

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THE UNIVERSITY OF NEWCASTLE AUSTRALIA

INFO6001: Database Management 1

Assignment 2

Project: Database design of UNIE Library System

(Logical Database Design)

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Preface

This is the requirements analysis, conceptual design and logical database design for the implementation of the database design of UNIE Library System. The requirements analysis includes the data requirements, transaction requirements and business rules. This will ensure that all the data required to be stored is identified, can be manipulated and is managed according to the library's business policies. The conceptual design includes an extended entity relationship diagram in EER which describes the required entities, their attributes and their relationships. It is further clarified with a data dictionary for the entities, attributes and also the relationships. This conceptual model was developed from the requirements analysis. From the conceptual model, a relational model is established and presented in the form of DBML, which is then normalised to BCNF, with a discussion of the steps to produce to logical design of the database. In this report, I first present reflection on Assigment 1 and comparision between the solution EER and the previously submitted EER in my Assignment 1 and described about the missed entities and attributes. Then represented the data requirements, transaction requirements and business rules using the solution EER. Also represented the mapping using the solution EER. And finally described why Normalisation is not necessary

Part1: Reflection on Assignment1:

This section presents necessary discussion to point out the differences between my submitted EER for assignment 1 and the solution EER. In my submitted EER for assignment 1, I haven't included the entities such as **MemberType, Privilege and Collection** which is mandatory. Also missed the few attributes that required for EER. Also, I have removed Publisher, Vendor, Item Author entities, which are not necessary for this design. Also I have missed few queries, relationships in first assignment, In the following discussions, the provided solution EER will be used in all the discussions.

Part 2: Requirements

Data Requirements

Entities with Attributes, Relationship & Cardinality:

1. Member

- **Description**: Stores information about library members. Each member has a unique ID and can have different membership types.
- **Attributes**:
    - memberId (Primary Key): Unique identifier for each member.
    - PIN: Authentication key for security purpose
    - name: Full name of the member.
    - dateOfBirth: Date of birth of member
    - address: Address of the member.
    - phoneNumber[1 to 4]: Phone number of the member(up to 4).
    - email: Email address of the member.
    - joinDate: Start date of the membership
    - dtMemberExpiry: Expiry date of the membership (if applicable).
    - status: Status of the member(Active or Inactive)

2. MemberType

- **Description**:
- **Attributes**:
    - typeId(Primary Key): Unique identifier for each type.
    - type: Type of the membership
    - description: brief description about membership type
    - notes: Additional information or rules about membership type
- **Relationships**:
    - **For** *(1..*)* → **Privilege** (Each membership type grants different privileges.)

3. Privilege

- **Description**: Stores information about membership type like student, staff, etc.,

- **Attributes**:
    - privilegeId(Primary Key): Unique identifier for each privilege.
    - name: Name of the privilege
    - description: brief description about privilege
    - loanPeriod: Maximum duration of a member can borrow an item
    - maxRenewals: Maximum allowed renewals of a member can extend the loan
    - maxItems: Maximum items a member can borrow.
    - maxHolds: Maximum items a member can reserve.

4. Item

- **Description**: Represents different types of items in the library, such as books, journals, and online databases.
- **Attributes**:
    - itemId (Primary Key): Unique identifier for each item.
    - title: Title of the item.
    - publisher: Publisher of the item.
    - author[0..10] - List of Authors for an Item
    - year: Year the item was published.
    - description: Brief description of the item.
    - notes: Additional details of the item.
    - subject: Subject or category of the item.
- **Subclasses**:
    - **Book**:
        - edition: Edition of the book.
        - contents: Table of contents or chapters.
        - summary: Brief summary of the book.
        - ISBN: International Standard Book Number.

- **Journal**:
    - frequency: Frequency of publication (e.g., Monthly, Yearly).
    - abbreviatedTitle: Abbreviated title of the journal.
    - mainSeries: Main series of the journal.
    - ISSN: International Standard Serial Number.
- **Online Database**:
    - contents: Description of the resourse. For example, papers, articles etc.,
    - releaseDate: Date the database was released.

5. Copy Of Item

- **Description**: Represents a physical or digital copy of an item.
- **Attributes**:
    - accessionId (Primary Key): Unique identifier for the copy of an Item.
    - dateAdded: Date the copy of an item was added to the library.
    - cost: Cost of the copy of an item.
- **Subclasses**:
    - **Physical Copy**:
        - barcode: Barcode of the physical copy.
        - status: Current status (e.g., Available, On Loan, On Hold, Damaged, Withdrawn, Lost).
    - **Digital Copy**:
        - URL: URL or link to access the digital copy.
        - format: File format (e.g., PDF, EPUB).
        - accessPeriod: Period for which the digital copy is accessible.
        - size: Size of the digital copy.
- **Relationships**:
    - **Located At** *()* → **Collection** (Each copy belongs to a collection.)

6. Collection:

- **Description**: Stores the information about collections.
- **Attributes**:
    - collectionId(Primary Key): Unique identifier for the collection.
    - physicalLocation: General storage location of the collection.
    - campus: Campus where exactly located to find the collection
    - building: Building name inside the campus
    - room: Room number inside the building.
    - shelf: Shelf name within the room
    -

6. Borrow

- **Description**: Tracks the borrowing of physical copies by members.
- **Attributes**:
    - borrowId(Primary Key): Unique identifier for the borrowing transaction.
    - borrowDate: Date the copy was borrowed.
    - dueDate: Due date for returning the copy.
    - returnDate: Date the copy was returned.
    - numOfRenewals: count of the borrow has been renewed
    - status: current status of the borrow(Ex: returned, active, overdue)

7. Hold

- **Description**: Tracks the holds placed on physical copies by members.
- **Attributes**:
    - holdId(Primary Key): Unique identifier for the hold request.
    - holdDate: Date the hold was placed.
    - holdTime:Time the hold was notified
    - status: Current status of the hold request(Ex: pending, cancelled, etc.,)

- comments: Additional remarks for the hold request

8. Suggestion

- **Description**: Tracks suggestions for new items made by members.
- **Attributes**:
  - suggestionId(Primary Key): Unique identifier for the suggestion.
  - typeOfItem: Type of the suggested item(Ex: Book, Journal, etc.,)
  - title: Title of the suggested item.
  - author[0..4]: Author of the suggested item(up to 4).
  - publisher: publisher of the suggested item.
  - edition: Editon of the suggested item.
  - description: Description of the suggested item.
  - Reason: Reason for aquisition of item.

Transaction Requirements:

Data Manuplation:

Insert Operations

- Add a new item (Book, Journal, or Online Database).
- Add a new item copy (Physical or Digital).
- Add a new collection.
- Register a new member.
- Record a new member type(Depends on member)
- Add a new borrowing transaction by member.
- Place a new hold request.
- Submit a new acquisition request.
- Add a new privilege.

Update Operations

- Update the status of a physical copy (e.g., change from "Available" to "On Loan").
- Update the details of an item (e.g., change the title or author).
- Update the membership details of a member.
- Update the due date of a borrowing transaction.
- Update the status of a hold request (e.g., from "Pending" to "Fulfilled").
- Update the details of a collection (e.g., change the physical location, shelf).
- Update the privileges of a membership type.

Delete Operations

- Remove an item from the library.
- Remove a copy of an item.
- Remove a member from the system.
- Remove a hold request.
- Remove a collection
- Remove a membership type
- Remove a privilege

Queries:

- Search for a book by title, author, or ISBN.
- Search for a journal by title, ISSN, or frequency
- Search for an online database by contents or release date.
- Search for members by name, email, or phone number.
- Search for collections by campus, building, or room.
- Check if a physical copy of an item is available for borrowing.
- Check the status of a digital copy (e.g., accessible or expired).
- Check the availability of a collection.
- View all items borrowed by a particular member.
- View all members who have borrowed a particular item.

- View all hold requests for a particular item.
- View all hold requests placed by a specific member.
- View all acquisition requests submitted by members.
- View the status of a specific acquisition request (e.g., pending, approved, rejected).
- View the borrowing history of a member.
- View the details of a specific collection.

Business Rules

Item Rules

- An item must belong to one of the following types: Book, Journal, or Online Database.
- The **Item** entity has three subclasses: Book, Journal, and Online Database.
- Each subclass has specific attributes that are not shared with the other subclasses.
    - Example: A Book has an ISBN, while a Journal has an ISSN. An Online Database has a releaseDate, which is not applicable to books or journals.

Member Rules

- Each member must belong to one membership type (e.g., Student, Staff, Faculty). The membership type determines the privileges granted to the member.
- Privileges (e.g., loan period, max renewals, max items) are assigned based on the member's membership type. These privileges are enforced during borrowing and hold transactions.

Collection Rules

- Each collection must have a physical location specified, including campus, building, room, and shelf. This ensures that items in the collection can be easily located.

Copy Rules

- A copy must be either a physical copy or a digital copy.
- A physical copy must have a barcode and status.
- A digital copy must have a URL, format, and access period.

- A physical copy status must be one of the following: Available, On Loan, On Hold, Damaged, Withdrawn, or Lost.

Digital Copy Rules

- Digital copies are accessible only for the specified access period. After the access period expires, the digital copy is no longer accessible to the member.

Member Rules

- Only members with an active membership can borrow items or place hold requests.
- A member can borrow a maximum of 5 items at a time. This limit is enforced based on the maxItems attribute in the Privilege entity.
- A member can place a maximum of 3 hold requests at a time. This limit is enforced based on the maxHolds attribute in the Privilege entity.

Borrowing Rules

- A physical copy can only be borrowed by one member at a time. The status of the copy must be updated to "On Loan" when borrowed.
- A physical copy can be borrowed for a maximum of 30 days. The dueDate is calculated based on the loanPeriod attribute in the Privilege entity.
- A member cannot borrow a copy that is already on hold by another member. The status of the copy must be checked before allowing a borrow transaction.

Renewal Rules

- A member can renew a borrowed item up to the maximum allowed renewals specified in their privileges.The numOfRenewals attribute in the Borrow entity tracks the number of renewals.

Notification Rules

- Members must be notified when an item they borrowed is overdue. Notifications are sent based on the dueDate and returnDate in the Borrow entity.

Hold Rules

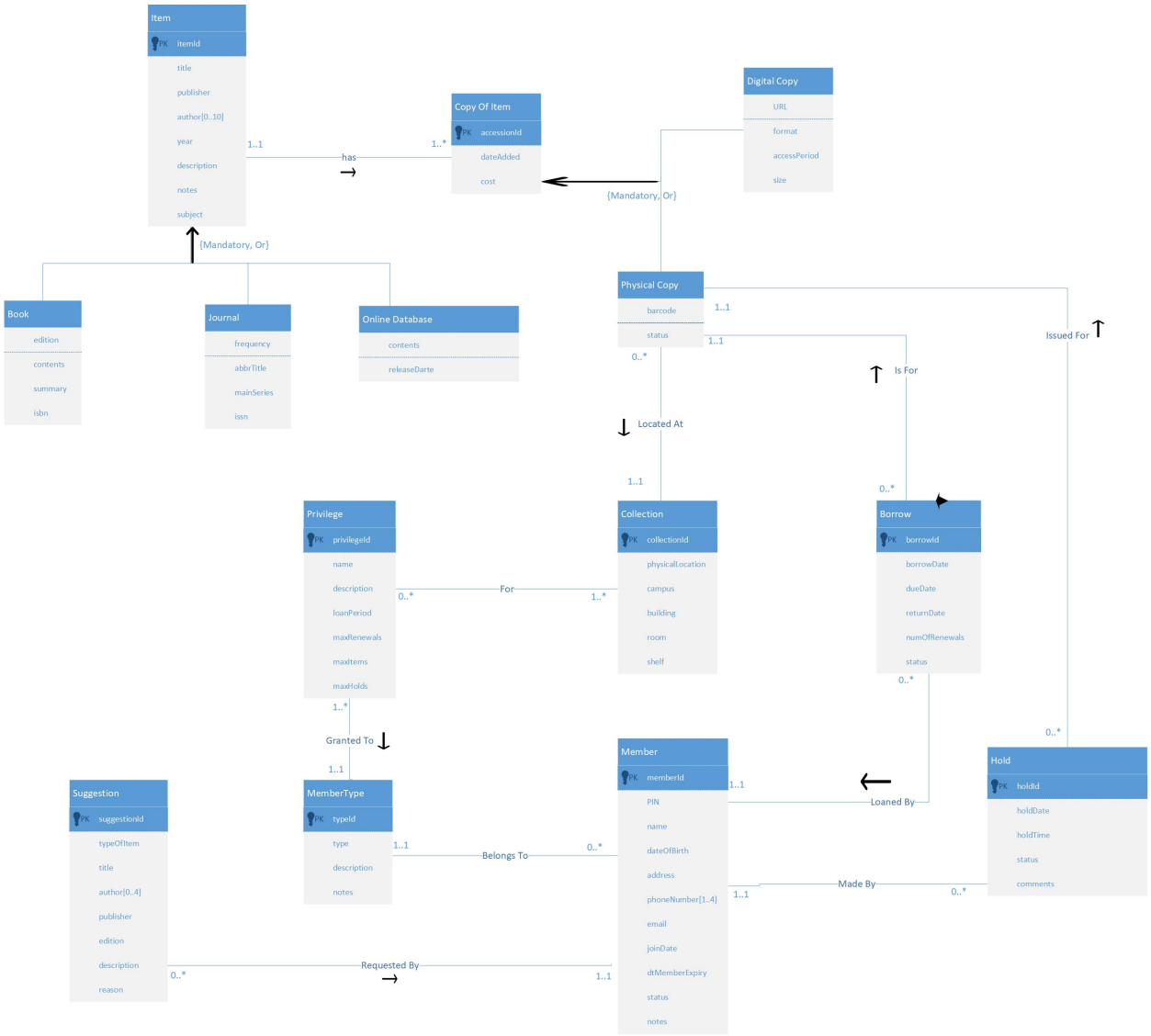
- A hold request is processed on a first-come, first-served basis. A member cannot borrow a copy that is already on hold by another member. The status of the copy must be checked before allowing a borrow transaction.
- A hold request expires after 7 days if not fulfilled. The status of the hold request is updated to "Expired" after the expiry period.

Acquisition Rules

- Members can suggest new items for acquisition.
- Acquisition requests must include a title, author, and subject.
- Acquisition requests are reviewed by library staff before approval.

Part3: EER Model with Data Dictionary:

EER Diagram:



Data Dictionary:

Entities & Attributes:

1.Member:

Column Name	Data Type	Constraints	Description
memberId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each member.
PIN	VARCHAR(50)	NOT NULL	Authentication key for security purposes.
name	VARCHAR(255)	NOT NULL	Full name of the member.
dateOfBirth	DATE	NOT NULL	Date of birth of the member.
address	VARCHAR(255)		Address of the member.
phoneNumber[1-4]	VARCHAR(20)		Phone number(s) of the member (up to 4).
email	VARCHAR(255)	UNIQUE	Email address of the member.
joinDate	DATE	NOT NULL	Start date of the membership.
dtMemberExpiry	DATE		Expiry date of the membership (if applicable).
status	VARCHAR(50)	CHECK (status IN ('Active', 'Inactive'))	Status of the member (Active or Inactive).

2.Member Type:

Column Name	Data Type	Constraints	Description
typeId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each membership type.
type	VARCHAR(50)	NOT NULL	Type of the membership (e.g., Student, Staff, Faculty).
description	TEXT		Brief description about the membership type.
notes	TEXT		Additional information or rules about the membership type.

3.Privilege:

Column Name	Data Type	Constraints	Description
privilegeId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each privilege.
name	VARCHAR(255)	NOT NULL	Name of the privilege.
description	TEXT		Brief description about the privilege.

loanPeriod	INT	NOT NULL	Maximum duration a member can borrow an item (in days).
maxRenewals	INT	NOT NULL	Maximum allowed renewals a member can extend the loan.
maxItems	INT	NOT NULL	Maximum items a member can borrow.
maxHolds	INT	NOT NULL	Maximum items a member can reserve.

4.Item:

Column Name	Data Type	Constraints	Description
itemId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each item.
title	VARCHAR(255)	NOT NULL	Title of the item.
publisher	VARCHAR(255)		Publisher of the item.
author[0..10]	VARCHAR(255)		List of authors for an item (up to 10).
year	INT		Year the item was published.
description	TEXT		Brief description of the item.
notes	TEXT		Additional details of the item.
subject	VARCHAR(255)		Subject or category of the item.

5.Copy Of Item:

Column Name	Data Type	Constraints	Description
accessionId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the copy of an item.
dateAdded	DATE	NOT NULL	Date the copy of an item was added to the library.
cost	DECIMAL(10,2)		Cost of the copy of an item.

6.Physical Copy(Subclass of Copy Of Item:

Column Name	Data Type	Constraints	Description
accessionId	INT	PRIMARY KEY	Links to the Copy Of Item entity.
barcode	VARCHAR(50)	UNIQUE	Barcode of the physical copy.
status	VARCHAR(50)	CHECK (status IN ('Available', 'On Loan', 'On	Current status of the physical copy.

		Hold', 'Damaged', 'Withdrawn', 'Lost'))	

7.Digital Copy(Subclass of Copy Of Item:

Column Name	Data Type	Constraints	Description
accessionId	INT	PRIMARY KEY	Links to the Copy Of Item entity.
URL	VARCHAR(255)	UNIQUE	URL or link to access the digital copy.
format	VARCHAR(50)		File format (e.g., PDF, EPUB).
accessPeriod	VARCHAR(50)		Period for which the digital copy is accessible.
size	DECIMAL(10,2)		Size of the digital copy.

8.Collection:

Column Name	Data Type	Constraints	Description
collectionId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the collection.
physicalLocation	VARCHAR(255)		General storage location of the collection.
campus	VARCHAR(255)		Campus where the collection is located.
building	VARCHAR(255)		Building name inside the campus.
room	VARCHAR(255)		Room number inside the building.
shelf	VARCHAR(255)		Shelf name within the room.

9.Borrow:

Column Name	Data Type	Constraints	Description
borrowId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the borrowing transaction.
borrowDate	DATE	NOT NULL	Date the copy was borrowed.
dueDate	DATE	NOT NULL	Due date for returning the copy.
returnDate	DATE		Date the copy was returned.
numOfRenewals	INT		Count of how many times the borrow has been renewed.

status	VARCHAR(50)	CHECK (status IN ('Returned', 'Active', 'Overdue'))	Current status of the borrow.

10.Hold:

Column Name	Data Type	Constraints	Description
holdId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the hold request.
holdDate	DATE	NOT NULL	Date the hold was placed.
holdTime	TIME		Time the hold was notified.
status	VARCHAR(50)	CHECK (status IN ('Pending', 'Cancelled', 'Fulfilled'))	Current status of the hold request.
comments	TEXT		Additional remarks for the hold request.

11.Suggestion:

Column Name	Data Type	Constraints	Description
suggestionId	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the suggestion.
typeOfItem	VARCHAR(50)		Type of the suggested item (e.g., Book, Journal).
title	VARCHAR(255)		Title of the suggested item.
author	VARCHAR(255)		Author(s) of the suggested item (up to 4).
publisher	VARCHAR(255)		Publisher of the suggested item.
edition	VARCHAR(50)		Edition of the suggested item.
description	TEXT		Description of the suggested item.
reason	TEXT		Reason for acquisition of the item.

Relationships:

- An **Item** (Book, Journal, or Online Database) can have **multiple copies** (1..*) but each copy belongs to only **one item** (1..1).
- An **Item** can be either a **Digital Copy** or a **Physical Copy** (Mandatory, Or).
- A **Copy of Item** can be either a **Digital Copy** or a **Physical Copy** (Mandatory, Or).

- A **Physical Copy** has attributes like barcode and status.
- A **Physical Copy** is **Located At** (1..1) a **Collection**.
- A **Collection** contains multiple **Physical Copies** (1..*).

- A **Member** belongs to **one MemberType** (1..1) but each **MemberType** can have multiple **Members** (0..*).
- A **Member** can have multiple **Privileges** (0..*).
- A **Privilege** is granted to multiple **Members** (0..*).
- A **Borrow** record is associated with a **Member** (0..* loaned by 1..1 Member).
- A **Borrow** is **for** an **Item** (0..*).
- A **Hold Request** is made by a **Member** (1..1) and is **for** an **Item** (0..*).
- A **New Item Request** is made by a **Member** (0..* requested by 1..1).

Part4: Mapping the EER to Relational Model:

Using the mapping rules, got the following relations for all entities in EER.

1. **Item:**

   Item (itemId, title, publisher, year, description, notes, subject)

   Primary Key: itemId

2. **Book:**

   Book (itemId, edition, contents, summary, isbn)

   Primary Key: itemId

   Foreign Key: itemId references Item(itemId) ON UPDATE CASCADE, ON DELETE CASCADE

3. **Jourrnal:**

   Journal (itemId, frequency, abbrTitle, mainSeries, issn)

   Primary Key: itemId

   Foreign Key: itemId references Item(itemId) ON UPDATE CASCADE, ON DELETE CASCADE

4. **Online Database:**

   OnlineDatabase (itemId, contents, releaseDate)

   Primary Key: itemId

Foreign Key: itemId references Item(itemId) ON UPDATE CASCADE, ON DELETE CASCADE

5. **CopyOfItem:**

CopyOfItem (accessionId, itemId, dateAdded, cost)

Primary Key: accessionId

Foreign Key: itemId references Item(itemId) ON UPDATE CASCADE, ON DELETE CASCADE

6. **PhysicalCopy:**

PhysicalCopy (accessionId, barcode, status)

Primary Key: accessionId

Foreign Key: accessionId references CopyOfItem(accessionId) ON UPDATE CASCADE, ON DELETE CASCADE

7. **DigitalCopy:**

DigitalCopy (accessionId, URL, format, accessPeriod, size, collectionId)

Primary Key: accessionId

Foreign Key: accessionId references CopyOfItem(accessionId) ON UPDATE CASCADE, ON DELETE CASCADE

Foreign Key: colectionId references Collection(collectionId) ON UPDATE CASCADE, ON DELETE CASCADE

8. **MemberType:**

MemberType (typeId, type, description, notes)

Primary Key: typeId

9. **Member:**

Member (memberId, PIN, name, dateOfBirth, address, phoneNumber, email, joinDate, dtMemberExpiry, status, typeId)

Primary Key: memberId

Foreign Key: typeId references MemberType(typeId) ON UPDATE CASCADE, ON DELETE CASCADE

10. Privilege:

Privilege (privilegeId, name, description, loanPeriod, maxRenewals, maxItems, maxHolds, typeId)

Primary Key: privilegeId

Foreign Key: typeId references MemberType(typeId) ON UPDATE CASCADE, ON DELETE CASCADE

11. Collection:

Collection (collectionId, physicalLocation, campus, building, room, shelf)

Primary Key: collectionId

12. Borrow:

Borrow (borrowId, memberId, accessionId, borrowDate, dueDate, returnDate, numOfRenewals, status)

Primary Key: borrowId

Foreign Keys:

memberId references Member(memberId) ON UPDATE CASCADE, ON DELETE CASCADE

accessionId references CopyOfItem(accessionId) ON UPDATE CASCADE, ON DELETE CASCADE

13. Hold:

Hold (holdId, memberId, accessionId, holdDate, holdTime, status, comments)

Primary Key: holdId

Foreign Keys:

memberId references Member(memberId) ON UPDATE CASCADE, ON DELETE CASCADE

accessionId references CopyOfItem(accessionId) ON UPDATE CASCADE, ON DELETE CASCADE

13. Suggestion:

Suggestion (suggestionId, memberId, title, author, publisher, edition, description, reason)

Primary Key: suggestionId

Foreign Key: memberId references Member(memberId) ON UPDATE CASCADE, ON DELETE CASCADE

Part 5: Normalising the Scheme up to BCNF:

By following the definitions of 1st Normal Form, 2nd Normal Form, 3rd Normal Form and BCNF, I found the functional dependencies in Member, Borrow, and Privilege tables.

1. **Member:**

   Member (memberId, PIN, name, dateOfBirth, address, phoneNumber, email, joinDate, dtMemberExpiry, status, typeId)

   I found the two functional dependencies:

   ***Functional Dependency-1:** memberId -> name, address, phoneNumber, email, joinDate, dtMemberExpiry, status, typeId

   ***Functional Dependency-2:** email-> name, address, phoneNumber(There is a partial dependancy).

   MemberId is a candidate key, but email is not candidate key, So we have to split the email, address, phoneNumber into seperate table.

   So final **BCNF Normalization** is:

   - Member (memberId, PIN, name, dateOfBirth, joinDate, dtMemberExpiry, status, typeId)

   Primary Key: memberId

   Foreign Key: typeId references MemberType(typeId) ON UPDATE CASCADE, ON DELETE CASCADE

- MemberContacts(email, phoneNumber, address, memberId)

Primary Key: email

Foreign Key: memberId references Member(memberId) ON UPDATE CASCADE, ON DELETE CASCADE

2. **Borrow:**

Borrow (borrowId, memberId, accessionId, borrowDate, dueDate, returnDate, numOfRenewals, status)

I found the two functional dependencies:

***Functional Dependency-1:** borrowId-> borrowDate, dueDate, returnDate, numOfRenewals, status, memberId, accessionId

***Functional Dependency-2:** accessionId -> dueDate, returnDate, numOfRenewals, status (There is a transitive dependancy).

borrowId is a candidate key, but accessionId is not candidate key, So we have to split the borrow data from accession data.

So final **BCNF Normalization** is:

- Borrow (borrowId, borrowDate, memberId)

Primary Key: borrowId

Foreign Keys:

memberId references Member(memberId) ON UPDATE CASCADE, ON DELETE CASCADE

- BorrowDetails(borrowDetailsId, borrowId, accessionId, dueDate, returnDate, numOfRenewals, status)

Primary Key: borrowDetailsId

Foriegn Keys: borrowId references Borrow(borrowId) ON UPDATE CASCADE, ON DELETE CASCADE

accessionId references CopyOfItem(accessionId) ON UPDATE CASCADE, ON DELETE CASCADE

3. **Privilege:**

Privilege (privilegeId, name, description, loanPeriod, maxRenewals, maxItems, maxHolds, typeId)

I found two functional dependancies.

***Functional Dependency-1:** privilegeId-> name, loanPeriod, maxRenewals, maxItems, maxHolds

***Functional Dependency-2:** name-> description(Transitive Dependancy)

previlegeId is a candidate key, but name is not candidate key, So we have to split the name, description into seperate table.

So final **BCNF Normalization** is:

- Privilege(privilegeId, name, description, loanPeriod, maxRenewals, maxItems, maxHolds, typeId)

Primary Key: privilegeId

Foreign Keys: typeId references MemberType(typeId) ON UPDATE CASCADE, ON DELETE CASCADE

Name references PrivilegeNew(name) ON UPDATE CASCADE, ON DELETE CASCADE

- PrivilegeNew (name, description)

Primary Key: name

Summary:

To summarize, the entities, attributes, and their details has been stated for UNIE Library System. The correct entities used in this Database design of UNIE Library System are Item, Book, Journal, Online Database, CopyOfItem, Physical Copy, Digital Copy, MemberType, Member, Privilege, Collection, Borrow, Hold and Suggestion. Each entities have attributes and each entity have unique identifiers. The relationships between the entities were defined. Also provided description, data type & length, nulls, multi-valued, derived, and default values for each attributes and primary key. Also implemented the EER model to Relational model and Normalization.